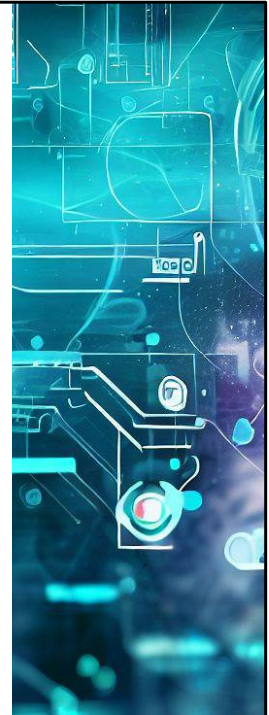


SC1006
Computer
Organisation and
Architecture
**Communication and
User Interface**

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This chapter deals with the more complex communication and user interfaces that a typical computer has.

I have grouped them here to explicitly indicate that we will only cover the overview and basics where these interfaces are concerned.

The questions related to this video will only appear in MCQs of the final exam.



Introduction

- This chapter introduce some of the common
 - Wired and wireless communication technology used in computers.
 - USB and HDMI
 - WiFi and Bluetooth
 - User interface devices used to transfer real world data.
 - Keyboard/Mouse
 - Capacitive Touch devices
 - Camera
 - Microphone
 - Display
 - Audio Speakers
- Only a general overview of the above will be discussed, technical details will not be covered.

- This chapter gives a basic introduction to the common interfaces seen on a computer system, particularly the Smartphone and the Lap Top.
- These includes the wired and wireless communication technology such as the USB, HDMI, Wifi and Bluetooth
- And other user interfaces such as
 - Key Board and Mouse
 - Capacitive Touch Devices
 - Camera
 - Microphone
 - Display, and
 - Audio Speakers
- Note again that we will only touch on the basics for the above. You can reference the lecture notes for the required depth you need to get into.

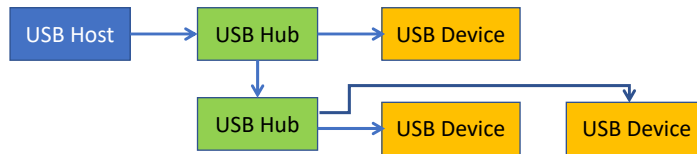
Universal Serial Bus (USB)



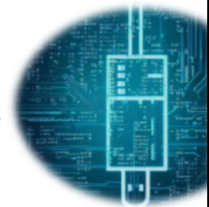
- **Serial Bus** designed to standardize connection of computer peripheral devices (Keyboard, Mouse, Printers, Disk Drives, Network Adapters, Digital Cameras etc).
- Effectively **replaced many interface buses** e.g. Serial Bus (COM Port), Parallel Port etc
- USB1.0 standard only supports a transfer rate of 12Mbps but has progressed over the years and the latest **USB3.2** standard supports up to **20Gbps** transfer.
- **4 basic pinout: VBUS** (+5V Power supply), **Data+/Data-** (Data pins) and **Ground** pin.

- First let's look at the most common external bus interface in the computer world today, USB.
- USB was first proposed to replace low speed peripherals such as the PC Serial COM Port, Keyboard/Mouse PS2 etc.
 - But it quickly become the de facto standard to interface to many other devices, e.g. Printers, Cameras, Disk Drives etc.
- The fast adoption is partly due to the continuous improvement in performance brought about by each newer version of the USB standard
 - It start with USB1.0 which only supports up to a data rate of 12Mbps,
 - But has progress over the years and the latest USB 3.2 standard is able to support up to 20Gbps transfer, sufficient to support real time video streaming.
- There are 4 basic pins in a USB interface
 - VBUS which is the power supply rail of the USB bus. This is also the pin that the USB devices can tap on for power supply to their system.
 - Data+/Data- pins which carries data information with a differential signal electrical interface.
 - Ground Pin for the VBUS supply.

USB Topology and interface



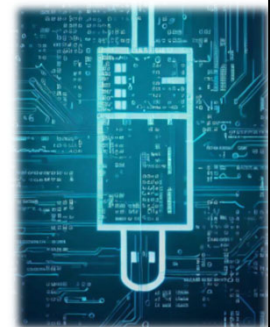
- USB used a **Tiered-Star Topology**, the center of the star is the USB host.
- All **data transactions** are **initiated by the USB Host** (typically resided on the computer). USB Peripherals (Mouse, Keyboard etc) can be connected directly to the **Host** or indirectly via the **USB Hub** devices.
- **Host will assign address to devices** connected to it to enable proper communication.
- All **data transaction** is **with respect to the USB Host**, i.e. data going into the Host is known as IN transaction and data going out of the Host is known as OUT transaction.
- **Power** to the devices can be **supplied by the Host or Hub**, known a **bus-powered**, or the device could have its **own power source (self-powered)**.



- USB uses a Tiered-Star Topology
 - The USB Host is at the center of the star, connecting to either USB Hub or USB Devices.
 - USB Hub can further connect to more USB devices, with itself at the center.
- All data transaction are initiated by the USB Host.
 - Each device is identified by its unique address which is assigned by the USB Host.
 - All data transaction is with respect to the Host, i.e. an IN transaction refers to data going into the Host while an OUT transaction transfer data out of the Host.
- As mentioned earlier, power for USB devices can be supplied via the VBUS.
 - If a device draws its power from VBUS, then it is known as a Bus-Powered Device.
 - Else, it is known as Self-Powered.

USB Enumeration and Device Class

- When the USB device is first connected to the Host, it will undergo an enumeration process where the device and the host will exchange information on their capability and requirement. E.g. a USB mouse when connected informs the Host its Vendor and Product ID, the device class it support, bandwidth required, etc.
- The host cannot communicate with the device until it is properly enumerated.
- Once all the device information is transferred to the Host, the host will check if it has the required device driver to support the USB device according to the USB device class that the device belongs to.
- There are many USB Device Classes, common ones are
 - Human Interface Device (HID) used in Mouse/Keyboard.
 - Communication Device Class (CDC) used to implement Virtual COM Port e.g. Arduino Board, MSP432 Launchpad used in the lab.
 - Mass Storage Class (MSC) used to interface to external USB HDD/SSD.
 - USB Audio Class used to stream audio to USB headset/Microphone.



- In order for USB host and Device to communicate with each other, they need to go through the USB enumeration process.
- When the USB device first connects to the Host, it will go through a process know as USB enumeration where the Host and Device will exchange information on their capability and requirement.
 - E.g. when a USB mouse is connected to the PC, it will inform the host its Vendor and Product ID, the polling rate it supports, the USB device class drivers it supports and needs, how many buttons it has, whether it is bus or self powered, etc.
- The host can only start communicating with the new USB devices after it has gone through the enumeration process successfully.
- Once the host understand the USB device's requirement, it'll check and load the required device drivers to support its operation.
- Device drivers are software doing the first level interface with the USB device's hardware.
- There are many different types of USB device class. Some of the more common ones are listed here.
 - Human Interface Device. HID in short. This device class is used in many of the user input devices such as Keyboard. Muse, touch pad etc.

- Communication Device Class., which is used to implement the Virtual COM Port used in many embedded system boards. E.g. Arduino, Raspberry Pi and the MSP432 Launchpad used in your lab sessions.
- Mass Storage Class device class which handles external mass storage devices such as the HDD, SSD and USB thumb drives.
- USB Audio Class which supports streaming audio via the USB bus. One example of such devices is the USB Headphone,

High-Definition Multimedia Interface (HDMI)

- HDMI is a [proprietary audio/video interface](#) for transmitting [uncompressed video data and compressed/uncompressed digital audio data](#) from a source device, such as a display controller in a computer, to a compatible HDMI receiver such as computer monitor, digital television, or digital audio device.
- In addition to transferring audio/video data, the [CEC \(Consumer Electronics Control\) capability allows HDMI devices to control each other](#) when necessary and allows the user to operate multiple devices with one handheld remote control device.
- Three commonly used type of HDMI connector type are shown in the figure on the right. Starting from the left: [Type D \(micro\)](#), [Type C \(Mini\)](#) and [Type A](#).



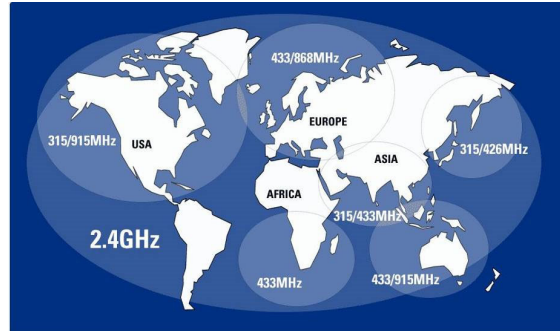
Source: <https://eocortex.com>

- The original HDMI v1.0/1.1 can only support a transfer rate of 3.96Gbps, allowing video format up to 1080p 60fps.
- The [latest HDMI v2.1 can achieve 42.6Gbps](#), allowing [8K resolution video](#) to be displayed.

- HDMI is a complex interface standard that is able to transfer high quality uncompressed audio and video data.
- In addition to the media data, it also support the CEC standards which allows HDMI devices to control each other and allow user to operate multiple HDMI devices with only one remote controller.
- Hardware interface wise, there are three commonly used HDMI connector, Type D, C and A. You can see these connectors in the photo, Type D is the micro version, which is the smallest connector on the left, this it followed by type C, which is the mini, and type A, which iis the regular HDMI connector.
- Similar to USB, HDMI has progressed through time, the first version HDMI1.0 can only support up to 3.96Gbps data transfer rate, but it has increased to 42.6Gbps in the HDMI v2.1. With this data transfer rate, the HDMI2.1 compliant device is able to transfer and display 8K high resolution video.

Industrial, Scientific and Medical (ISM) RF Band

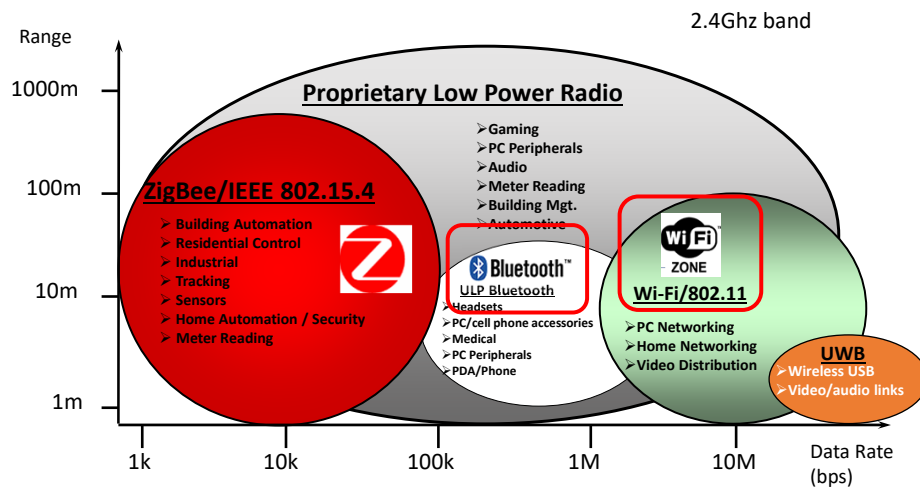
- A range of “Royalty Free” Radio Frequency Bandwidth
- Some are applicable world wide while some are restricted to certain geographical regions.
- The two commonly known world wide ISM bands are 2.4Ghz and 5.8Ghz.



Source: <https://predictabledesigns.com/most-important-decision-when-creating-wireless-product/>

- If you haven't realised, the use of various frequency bands in the air to transmit data is typically not free, i.e. you need to pay the government or some organisation in order to transmit data in most frequency bands.
 - E.g. the telecommunication companies like Singtel, Starbhub, and M1 has to pay the Singapore government licensing fees in order to transmit data via the 3G/4G cellular bands.
- There are however, some frequency bands that are free for all to use.
 - These are known as ISM Bands, which stands for Industrial, Scientific and Medical RF Band.
 - ISM frequencies varies with the geographical regions, but there are two frequency bands that are standard world wide.
 - 2.4Ghz and 5.8Ghz bands.
 - Incidentally, these are also the two bands that are used by the Wifi transmission standards.

Wireless Standards in ISM



Source: Texas Instruments

Pic Source: Texas Instruments

- One of the most commonly used ISM band is the 2.4Ghz band, this slides shows some of the RF standard that uses this frequency band for data transmission.
- We will be touching on two of these standards, which is the Wifi and Bluetooth.

WiFi

- A family of wireless networking technologies, based on the IEEE 802.11 family of standards, commonly known as “Wireless LAN”.
- Operates in the 2.4Ghz and 5.8Ghz RF range.
- Two common topologies: Infrastructure and Adhoc.
- Infrastructure Mode
 - Uses Star topology, at the center of the network is an Access Point or Router, connected to devices at the end.
- Adhoc Mode
 - Peer to Peer connection
- Transmission Range generally between 20m to 150m, factors affecting the range include transmission frequency, transmission power and interference.
- Transfer rate of up to ~10Gbps for the latest 802.11ax (WiFi 6).



- Wifi is based on the IEEE802.11 family of wireless transmission standard.
- It initially only utilise the 2.4Ghz RF Band but has progress in recent year to use both 2.4Ghz and 5.8Ghz bands.
- Two common topologies used in Wifi are the Infrastructure and Adhoc modes.
- Infrastructure mode use a Star Topology.
 - The Access Point or Router is at the center and terminals such as you phones, tablets and lap tops are the end devices.
- Adhoc mode is a peer to peer connection between two wifi enabled devices.
 - One example is the Wifi Direct technology.
- Transmission range for wifi is generally around 20 to 150m.
 - Factors that affects transmission range includes
 - Transmission frequency
 - Transmission power
 - RF interference
 - Wifi data transfer rate of 10Gbps is possible with the latest 802.11ax, also call Wifi6.

Bluetooth



- Mainly used for low data rate wireless transmission with focus on low power consumption.
- Example: bluetooth headset, smart wearables, Bluetooth mouse etc.
- Operates in 2.4Ghz range
- Transmission range up to 100m but typically kept to 10-20m to keep power consumption low. Factors affecting transmission range is similar to that of WiFi.
- Star topology
- Transfer rate in order of Kbps and Mbps.
- The Bluetooth standard defined two different Bluetooth protocols: Bluetooth Classic (BT Classic) and Bluetooth Low Energy (BLE).
- These are based on two completely different network protocols and are not compatible. But most Bluetooth Hosts today are “Dual Mode” host so is able to connect to both BT Classic and BLE devices.

- Bluetooth transmission standard targets low power applications, compared to Wifi transmission standard.
 - the transmission rate is much lower than Wifi as well.
- Example of Bluetooth applications includes headset, keyboard, mouse, smart wearables, etc.
- Bluetooth devices operates in the 2.4Ghz band
 - Transmission range can be up to 100m but is typically kept to 10-20m to keep the power consumption low.
 - Factors affecting the transmission range are similar to Wifi; in fact, the factors discussed are applicable to all RF transmission standards.
- Bluetooth also used a star topology
 - The center of the network is known as a Bluetooth Central, the end device known as device or peripheral.
- Transfer rate are typically lower than wifi and is in the order of Kbps and Mbps; in fact typically more Kbps than Mbps.
- Note that the Bluetooth standard defines two completely different protocols
 - One is the protocol defined in the original Bluetooth standard; this is known as the Bluetooth Classic.
 - The other is a new protocol adopted later to further improve the power

consumption; this is known as Bluetooth Low Energy. BLE in short.

- These two protocols are not compatible with each other.
- But most Bluetooth Host on Smart Phone, Tablets or Lap Top these days are “Dual Mode” Host and is able to talk to both BT Classic and BLE devices.

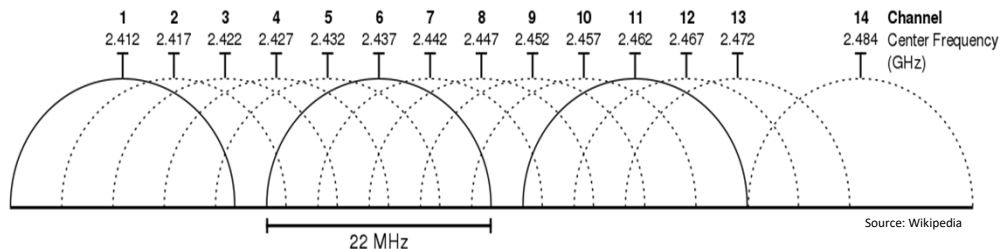


Factors affecting transmission range

- Transmission power
 - Transmission range increase as transmission power increase.
- Transmission frequency
 - Higher frequency signal experience higher attenuation when propagating through the air or other medium.
 - All things equal, higher frequency signal has lower range than lower frequency signal.
- Interference
 - Many commonly adopted standards such as Wifi and Bluetooth works in the same 2.4Ghz ISM band. Their transmission will interfere with each other.
 - The closer the transmitter is to each other, the stronger the interference.
 - Even the micro oven in your kitchen operates in the 2.4Ghz range.

- We will go into a little detail on the factors affecting wireless transmission range and the mitigating methods, in the next two slides.
- As mentioned in previous slides, the discussion here is applicable to all RF transmission standards.
- First factor that affects the transmission range is the transmission power.
 - Larger transmission power will give rise to larger range.
- Rate of attenuation increases if the transmission frequency increases.
 - i.e. a 2.4Ghz RF signal will suffer less attenuation compared to a 5.8Ghz signal.
- One of the key factors that affect transmission range is the interference from other transmission sources.
 - This is especially so for 2.4Ghz which is used by many RF transmission standards.
 - E.g. Wifi, Bluetooth, Cordless Phones, Baby Monitor, etc
 - Even your Microwave Oven uses 2.4Ghz!
 - In general, the closer the transmitter is from each other, the stronger the interference.

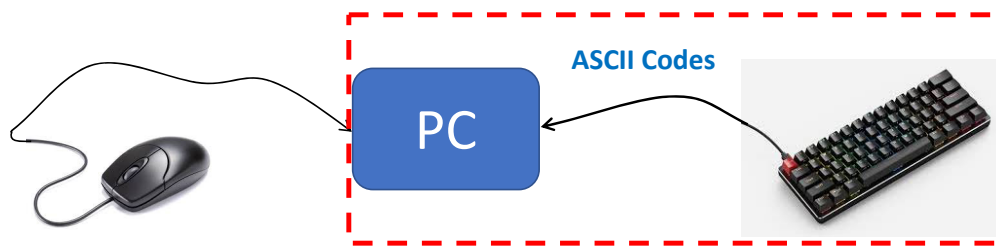
Mitigating Interference (some methods)



- “2.4Ghz ISM band” really consist of a [band of frequencies between 2.4 and 2.5Ghz](#). Similarly, “5.8Ghz band” also span across a certain frequency range.
- Figure above shows the Wifi standard dividing the given frequency band into sub-bands known as [channels](#).
- One common way to mitigate Wifi interference is to [select different channels](#) to use from your neighbours.
- Another common way is to use [frequency hopping](#), i.e. to constantly hop from one channel to another so that transmission will eventually succeed. Bluetooth uses frequency hopping.
- There are [other methodologies and techniques](#) employed to further mitigate interference between transmitters.

- As interference is one of the most common factor affecting the RF transmission range, a lot of effort is put in to mitigate the effect of interference.
- Before we go into the actual detail, it is useful to know that the so call 2.4Ghz that wifi works on is actually a band of frequencies rather than exactly 2.4Ghz.
- The 2.4Ghz band that wifi or Bluetooth works on is shown in the diagram in the slide.
 - It is roughly between 2.4 to 2.5Ghz.
 - The 802.11 standards that Wifi and Bluetooth use divides this frequency band into multiple sub-bands known as channels.
 - At any point in time, the wifi or Bluetooth will use one or more of these bands for transmission.
- So, one way of mitigation is to have different device use a different channel or channels for transmission.
- Another way, which is used in Bluetooth, is to hop from one channel to the other periodically, so as to reduce the chances of colliding into each other’s operating band. This is known as Frequency Hopping.
- There are other methodologies and techniques employed to further enhance the transmission,.

Keyboard and Mouse

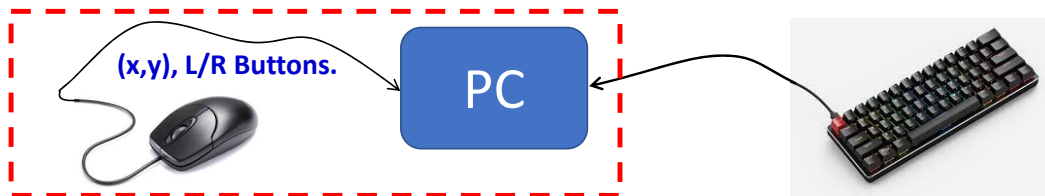


- **Keyboard**

- User input device for transmitting characters (alphabets, numbers, special symbols etc).
- Has a small micro-controller on board to detect the key press and send their corresponding **ASCII codes** to the computer.
- Connection to PC is via wired or wireless means.
- **Wired** keyboard today mainly uses **USB interface**, keyboard will be enumerated as a HID Keyboard device.
- **Wireless** keyboard mainly operate in the **2.4Ghz ISM**, technology could be Bluetooth or proprietary 2.4Ghz RF protocol.

- Keyboard and Mouse are the two important user input devices in a computer system.
- Both devices have a micro-controller in them that sent the main processor in the computer information based on the keys, buttons and mouse movement status.
- For Key Board, note that a coded form of the key status is sent rather than the actual value.
 - E.g. when the Key Pad '1' is pressed, the ASCII code of '1' is sent to the main processor.
- Connection to the PC can be wired or wireless
 - Wired Keyboard typically use the USB interface
 - The devices will be enumerated as a HID device class under USB standard.
- Wireless KeyBoard mainly operate in 2.4Ghz ISM band, it could either be using Bluetooth technology or some proprietary protocol designed by the vendors.

Keyboard and Mouse



• Mouse

- User **pointing device**, tracks its position by mechanical or optical means.
- Information reported are the **(x,y) coordinates and the left/right mouse button**.
- More information may be reported for mouse with more advanced features.
- Information reported back to PC periodically and is known as the **scan/polling rate** of the mouse.
- Typical mouse has a scan rate of **125Hz**, gaming mouse scan rate could be up to **1000Hz** or higher. Higher scan rate typically implies better response.
- Another parameter is the **Dot-Per-Inch (DPI)**, which measures how fine the mouse could track the physical movement, higher DPI implies **higher sensitivity to small physical mouse movements**.
- Connection to PC utilise similar technology as Keyboard.

- Mouse is the main pointing device use in a computer system.
- It tracks the cursor movement on the screen and is the key enabler for icon-based GUI.
- The tracking is done via mechanical or optical means.
 - For mechanical, there is a ball at the base of the mouse that will rotate as user slide the mouse, the ball will in turn move two rollers that track the X and Y axis movement.
 - For optical, the information transferred between the transmitter-receiver pair underneath the mouse tracks the movement of the mouse.
- Basic information reported by the mouse are the X/Y coordinates and the left/right mouse button press.
 - There may be more information reported for more advanced mouse, e.g. scrolling, panning and customised key codes etc.
- How fast these information gets reported back to the PC is known as the polling rate of the mouse.
 - Typical mouse has a polling rate of 125Hz but gaming mouse could have polling rate of up to 1000Hz or higher.
 - Higher polling rate mean faster response.
- Another Parameter is the DPI, which stands for Dot-Per-Inch.
 - This measures how fine the mouse is able to track the physical movement.
 - Higher DPI means higher sensitivity to small physical movement.
- Connection to PC is either wired or wireless and technology used is similar to the KeyBoard,

ASCII Table

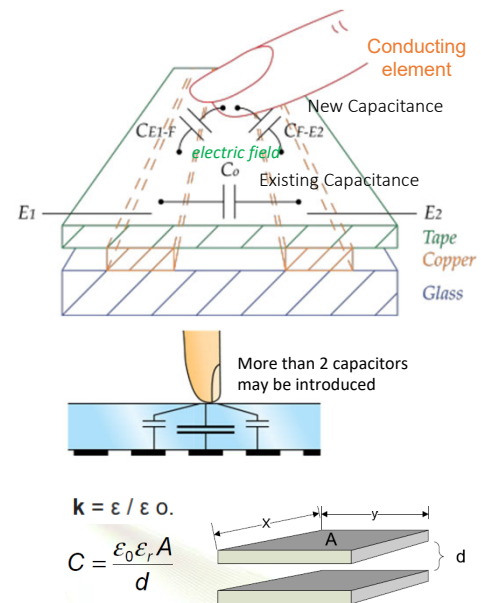
MS LS	0	1	2	3	4	5	6	7
0	NUL	DLE	SP	0	@	P	'	p
1	SOH	DC1	!	1	A	Q	a	q
2	STX	DC2	"	2	B	R	b	r
3	ETX	DC3	#	3	C	S	c	s
4	EOT	DC4	\$	4	D	T	d	t
5	ENQ	NAK	%	5	E	U	e	u
6	ACK	SYN	&	6	F	V	f	v
7	BEL	ETB	'	7	G	W	g	w
8	BS	CAN	(	8	H	X	h	x
9	HT	EM	)	9	I	Y	i	y
A	LF	SUB	*	:	J	Z	j	z
B	VT	ESC	+	;	K	[	k	{
C	FF	FS	,	<	L	\	l	
D	CR	GS	-	=	M	]	m	}
E	SO	RS	.	>	N	^	n	
F	SI	US	/	?	O	_	o	DEL

ASCII Character Set (7-Bit Code)

- This is the ASCII table which has been introduced to you previously during the first half.
- Keypad '1' has an ASCII code = 0x31 (MS = 3, LS =1 in the table).
- So 0x31 is sent to the main processor in the computer when the keypad '1' on the keyboard is pressed.

Capacitive Touch Interface

- A **capacitive touch pad/button** is seen as a **capacitor**.
- When a conductive element is present, e.g. finger, the **effective capacitance of the setup increases** (C_0 and combination of C_{E1-F} and C_{F-E2} in parallel)
- Capacitance is also affected by any **dielectric** e.g. gloves, plastics, liquid between finger & pad.
- Capacitance is **directly proportional to dielectric constant (k)** and air typically has a smaller dielectric constant (~ 1), compared to all other materials (>1).
- That's why your phone touch screen, which is typically capacitive touch based, don't work as well if you have a glove on or the screen is wet.
- **Calibration** is done under assumption of human finger touch; A **processor** is used to process the information to derive the **exact position** of contact.

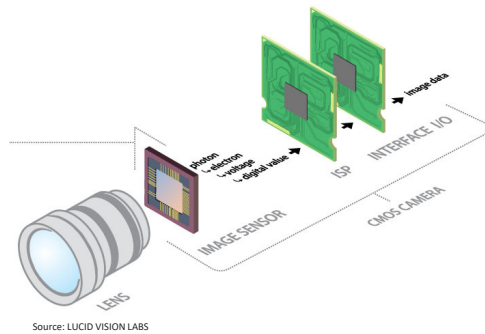


- Capacitive touch technology is widely used to implement touch button, pads and screen for smartphones, tablets and laptops.
- The working principles behind this technology is the change in capacitance of the hardware setup in the presence of human finger.
- The diagram on the right illustrates this principle
 - The copy electrodes are seen as a capacitor, C_0 , to the processor connected to it.
 - When an electrically conductive element such as the human finger is present in the vicinity, the effective capacitance of the setup changes.
 - The diagram illustrates the effect of putting a finger near a capacitive touch button, additional Capacitors (C_{E1-F} and C_{F-E2}) are formed between the fingers and the copper electrodes.
 - This change result in a change in electrical behaviour which can be picked up by the processor.
- Changes in capacitance is also affected by the dielectric of the capacitor.
 - This dielectric could be the glove, plastic sheet or a film of water between the finger and the screen.
 - Capacitance is directly proportional to the dielectric constant of the dielectric material and air typically has a lower dielectric constant than

most materials.

- This is why your phone's touch screen doesn't work or is not as responsive if your hand is wet or you wore your gloves.
- As the change in capacitance for a setup differs with different conductive medium interacting, calibration is needed to ensure a correct and reliable result.
- A **processor** is used to process the information to derive the **exact position** of contact.

Camera



- Typical camera on phones or PC today uses **CMOS** camera module, consisting of the sub-modules shown in the figure above.
- Sub-modules: **Lens, Image Sensor, Image Signal Processor and Interface I/O.**

- Most phones, tablets and laptop these days are equipped with one or more cameras.
- There are two main types of camera sensors used in the market, CCD and CMOS.
 - Most products with thin mechanical profile use the CMOS camera.
 - The camera module consist of the following main components
 - Lens
 - Image Sensor
 - Image Signal Processor
 - Interface I/O

Camera Sub-Modules

- Lens
 - Optical lens to focus the real world objects onto the image sensor.
- Image Sensors
 - Mostly CMOS based these days.
 - transduce photons (light information) to electrical signals (analog).
 - Analog-to-Digital conversion of the analog electrical signal to its digital equivalent for processing in the ISP.
- Image Signal Processor (ISP)
 - Processor designed to specifically handle image processing of collected image data from the sensor.
 - Processing done include Auto Focus, Auto Exposure, Auto White Balance, image format conversion, image post-processing/compression etc.
 - Common image formats are YUV (luminance and chrominance) and RGB (Red, Green, Blue).
- Interface I/O
 - Re-formatting of image data from the ISP for delivery to the host processor.



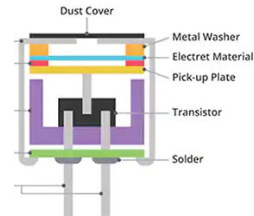
- Camera Lens is a transparent medium used to focus the real world objects onto the image sensor.
- Image Sensors are mostly CMOS based
 - CMOS stands for Complementary Metal Oxide Semiconductor, which is a semiconductor process technology based on MOSFET transistors.
 - A sensor IC consists of light sensors that transduce photons to analog electrical signals
 - These analog signals are then passed into ADC to be converted to digital signal that can be processed by the digital processor.
- The processor on the camera module is known as the Image Signal Processor, ISP in short.
 - Digital data of images detected by the sensor IC is sent to the ISP for further processing. Some of these processing are
 - Auto Focus, Auto Exposure and Auto White Balance, commonly known as '3A'.
 - Conversion between different image format, e.g. YUV, RGB etc.
 - Image post processing such as noise reduction and lens distortion adjustment etc
 - Image or Video compression to various formats such as JPEG, H.264

etc.

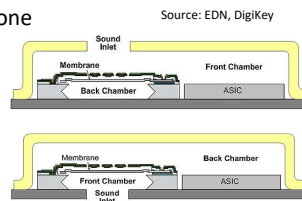
- The camera module needs to talk to the main processor in the computer system and that it done via the interface I/O controller on the module.
 - The I/O controller encapsulate the image and video data in a format that the processor support.

Microphones

ECM



MEMS Microphone



- Two of the most popular types of microphones are **micro-electro-mechanical system (MEMS) microphones** and **electret condenser microphones (ECM)**.
- Both MEMS and ECM microphones use the **variation in capacitance** when the **diaphragm is displaced by the sound pressure** to transduce sound wave to electrical signals.

$$k = \epsilon / \epsilon_0$$

$$C = \frac{\epsilon_0 \epsilon_r A}{d}$$

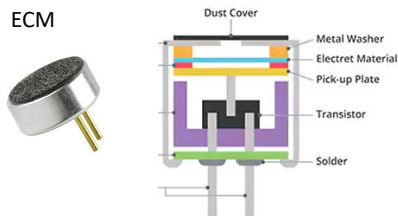
$d \text{ change} \rightarrow C \text{ change} \rightarrow V \text{ change}$

- The transduced signal is analog in nature but an ADC could be added to enable a digital output.

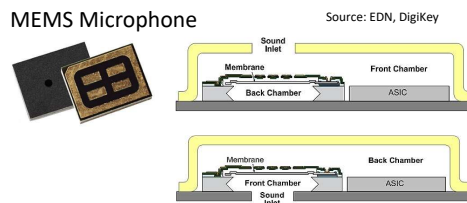
- Microphones are present in almost all computer systems that support voice communications.
- Two of the most popular types of microphone are the MEMS and ECM.
- Operating principles of both types of microphone are similar. Both have a diaphragm that can be displaced by sound waves, movement of the diaphragm in turn changes the capacitance of the setup, as in the figure here, $d \text{ change} \rightarrow C \text{ change}$, and these capacitance changes can be picked up by the electrical circuits connected and sent to the processor for further processing.
- The transduced signal is analog but an ADC can be included in the microphone to enable a digital output.

Microphones

ECM



MEMS Microphone



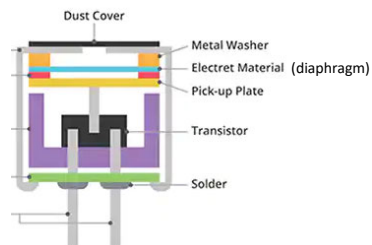
Source: EDN, DigiKey

- Difference is the in ECM, the electrical charges needed to measure the change in capacitance is provided by the charges stored on the electret.
- In MEMS Microphone case, the electrical charges needed is provided by a charge pump instead.



- The difference between MEMS and ECM is that
 - in the ECM, a fixed electrical charges needed to measure the change in capacitance is stored permanently in the electret. The electret here forms one of the capacitor plate.
 - While for MEMS microphone , a fixed electrical charges for the capacitor plate is provided by a charge pump circuit instead.

ECM Microphone

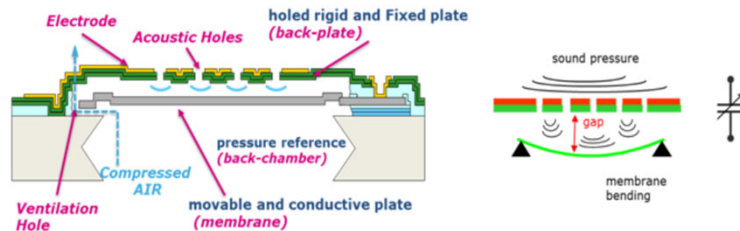


- In an ECM, the electret diaphragm is a material with a fixed surface charge that's placed near a conductive plate
- A capacitor is created with the air gap forming the dielectric.
- Sound pressure waves moving the electret diaphragm cause the value of the capacitance to change, causing voltage across the capacitor to vary, $V = Q/C$ (Q = a fixed charge).



- In the ECM microphone, the two capacitor plates are formed by
 - an electret, which stores some excess charges permanently and form the movable plate of the capacitor.
 - A fixed conductive plate connected to the rest of the circuitry of the microphone.
- The pressure created by the sound wave causes the electret to move, changing the gap of the capacitor (e.g., d in the previous slide) and hence the capacitance.
- Advantage of the ECM microphone is that its performance is more robust under scenario of fluctuation in supply voltage, because the charges embedded on the electret are permanent and not affected by variation in supply voltage.

MEMS Microphone

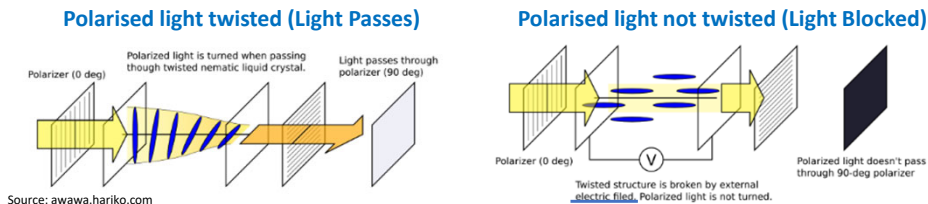


- MEMS microphone basically is an acoustic transducer.
- Transduction principle is the coupled capacity change between a fixed plate (back-plate) and a movable plate (membrane)
- The capacitive change is caused by the sound, passing through the acoustic holes, that moves the membrane modulating the air gap comprised between the two conductive plates

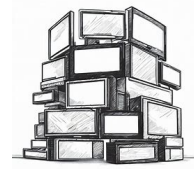


- MEMS microphone also employs the design of one fixed and one movable capacitor plate.
 - The back plate is charged up by an internal charge pump.
 - The pressure from sound wave pass through the acoustic holes and move the movable conductive plate, this changes the capacitor plate gap (again, d in the previous slide) and therefor the capacitance of the setup.
- MEMS microphones are more popular these days because compared to ECM microphone, MEMS microphone
 - Has a much smaller form factor
 - Better signal to noise ratio, i.e. better recording quality.

Liquid Crystal Display (LCD) Basics



- **Liquid crystal** is an organic substance that has both a liquid form and a crystal molecular structure. The rod-shaped molecules are **able to kept their order in a particular direction**.
- An **electric field** can be used to **control the molecules orientation**, according to pixel value (brightness) of an image.
- Depending on their orientation, these molecules is able to **twist the light passing through them**.
- The **amount of light** that is able to pass through the polariser **depends on its orientation with respect to the polariser**.
- This result in **light passing, partial passing or being blocked** from user point of view.

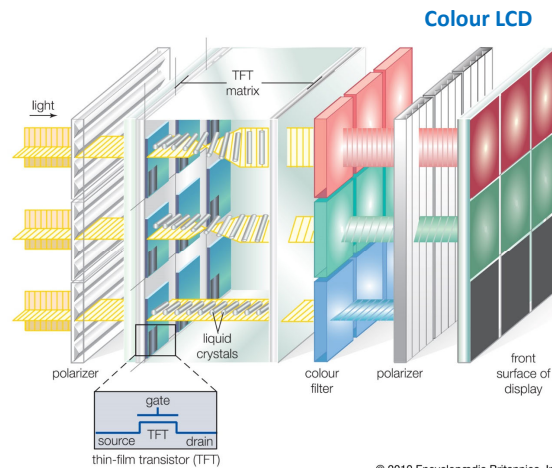


- The discussion here is the basic principle behind LCD, and the LCD technology you see in the market has various enhancement applied over this basic design.
- Liquid crystal is a substance that has both liquid and crystal property.
 - The rod-shaped molecules are able to keep their order in a particular direction although they are in a liquid state.
 - And the orientation of these molecules can be controlled with an electric field, according to pixel value (brightness) of an image.
- These molecules are able to twist the orientation property of the light passing through them.
- Next we need to introduce another important module of the LCD panel, the light polariser.
 - A light polariser allows light that has the same orientation as the polariser to pass through
 - Lights that has orientation that are 90 degree with respect to the polariser will be blocked completely
 - Any other orientation will have partial component of the light passing
- With a setup consisting of two polariser oriented 90 degrees from each other, as shown in the two figures, we will be able to control the amount of light passing through the setup by applying electric field to change the orientation of the LCD

molecules.

- If the LCD molecules twist the light by 90 degrees, then all the light will pass through the second polariser, as the figure on the left.
- If the LCD molecules keep the orientation of the light unchanged, then the light will be blocked by the second polariser, as the figure on the right.
- More general cases are between these two extreme ones: partial passing of light.
- In this way, we can display any pixel in an image.

Colour LCD

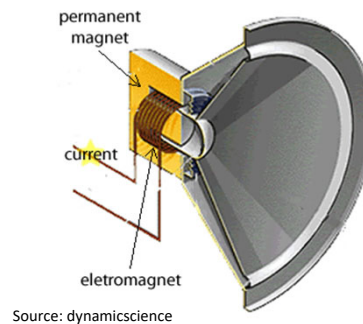
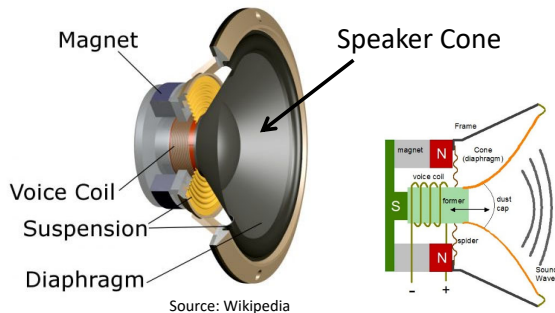


- Process for colour LCD is similar to that discussed in previous slide.
- Each pixel is associated with three colour filters to project the RGB colours.
- A varying orientation of molecules in the liquid crystal suspension: varies the amount of light allowed to pass through to the colour filter, thereby changing the colour picture on the display screen.



- Applying the LCD operating principles we learnt in the previous slide for the case of a colour LCD.
- The primary colours are Red, Green and Blue.
 - Any colour can be derived by mixing R, G, B in different proportions.
- Using this concept, we can have three light sources, one each for R, G and B, enabled by passing the light source through the respective colour filter.
- Each of these colour component will pass through the first polariser, LCD and second polariser setup that we discussed in previous slide.
 - By varying the electric field of each of the light source's LCD suspension, we can vary the amount of light for each colour component and hence derive the intended colour.

Audio Speaker



- The device **transduces electrical energy to sound energy**.
- The **speaker cone** vibrates, pushing and pulling the air to create sound waves.
- The conversion **from electrical to mechanical energy** occurs through an electromagnetic coil and magnet combination attached to the cone. This coil **moves the speaker cone back and forth** as its electromagnetic field changes with the electrical current passing through it, **converting the mechanical energy to sound energy**.



- Audio speaker transduces electrical energy to sound energy.
- The speaker cone is a moveable structure attached to an electrical coil called the Voice Coil.
- The voice coil is surrounded by a permanent magnet and moves according to the amount of electric current passing through coil.
- This in turn causes the speaker cone to move back and forth to create the sound waves which is heard as sound by the human ear.
- This same principle is used in headphones as well.

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